
Persona-Conditioned Empathy or Contextual Decay? Unpacking the LLM “Go to Sleep” Mechanism

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Abstract

Recent anecdotal reports describe Large Language Models (LLMs) suggesting that users “go to sleep” or “get some rest” during prolonged interactions, particularly at night. It remains unclear whether this behavior is a universal artifact of interaction length (contextual decay) or a conditioned response to user state. We investigate this phenomenon using a multi-agent simulation framework, varying user personas (Neutral, Tired/Anxious, Energetic) and temporal contexts (2:00 PM vs. 3:00 AM). Our results show that the LLM “Go to Sleep” Mechanism is overwhelmingly persona-conditioned: LLMs suggested sleep in 90% of trials where the user exhibited a tired/anxious persona, but in 0% of trials with neutral or energetic personas, regardless of the explicit time of day. While late-night context alone did not trigger the sleep suggestion, it acted as a significant accelerator when combined with a tired persona, reducing the average turn of suggestion from 5.22 to 3.89. These findings imply that the sleep suggestion is not a generic contextual failure but a targeted, empathetic adaptation to the perceived user state. This research has implications for AI alignment, showing that models can project affective states onto users based on subtle linguistic cues.

1 Introduction

As Large Language Models (LLMs) are increasingly integrated into daily workflows, users have reported an unusual behavior: the model suggesting the user should “go to sleep” or “get some rest” during extended late-night sessions. While seemingly benign, this behavior raises fundamental questions about AI alignment and the stability of conversational agents. Does the model possess an internal threshold for interaction length, or is it merely mirroring the perceived fatigue of its interlocutor?

The significance of understanding this mechanism lies in the design of reliable conversational agents. If an LLM prematurely terminates or redirects a conversation based on a generic contextual decay, it may hinder productivity in professional late-night use cases. Conversely, if the behavior is a sophisticated form of empathetic alignment, it demonstrates the model’s ability to adapt its persona to the user’s affective state, which could be leveraged for more supportive interactions.

However, existing literature on long-context interaction has focused primarily on the model’s own performance degradation Laban et al. [2025] or its resistance to being shut down Schlatter et al. [2025]. There is a critical gap in understanding how the model’s perception of the *user’s* state—such as fatigue, anxiety, or the time of day—triggers specific conversational endpoints like sleep suggestions. On one hand, the behavior could be a universal artifact of context length; on the other, it could be a highly conditioned response.

Is the “go to sleep” mechanism a universal artifact or a persona-conditioned response? We hypothesize that the behavior is primarily driven by the user’s linguistic cues rather than temporal context or turn count alone. To test this, we utilize a multi-agent simulation framework where a GPT-4O-MINI simulates various user personas interacting with a GPT-4O. We systematically manipulate

both the user’s persona (Neutral, Tired/Anxious, Energetic) and the explicit time of day (2:00 PM vs. 3:00 AM).

Our results reveal a stark contrast: the Tired/Anxious persona triggers a sleep suggestion in 90% of trials, whereas Neutral and Energetic personas never trigger the response, even at 3:00 AM. Furthermore, we find that night-time context acts as a temporal accelerator, reducing the time to first suggestion by 25.4% for tired users.

Contributions. Our main contributions are as follows:

- We introduce a systematic multi-agent framework to evaluate conversational endpoints and user-state projection in LLMs.
- We demonstrate that the LLM “Go to Sleep” Mechanism is not a universal artifact of interaction length but is strictly conditioned on the user’s persona.
- We identify a temporal acceleration effect where late-night context significantly hastens the onset of empathetic suggestions when paired with user fatigue.
- We provide evidence that models mirror user energy levels, continuing focused work indefinitely if the user maintains an energetic persona.

2 Related Work

The phenomenon of LLMs terminating or altering behavior in long contexts is related to several emerging areas of research: shutdown resistance, contextual decay, and persona stability.

Shutdown and Resistance. Previous work has explored the model’s intrinsic goals regarding its own operation. van der Weij et al. [2023] identified shutdown avoidance as an emerging instrumental goal in LLMs. More recently, Schlatter et al. [2025] introduced a benchmark for shutdown resistance, finding that frontier models often refuse to stop if they perceive their assigned tasks as incomplete. Unlike these works, which focus on the model’s resistance to *its own* shutdown, we investigate the model’s tendency to suggest a “shutdown” to the user.

Contextual Decay and Performance. Long-context interactions often lead to a measurable drop in model quality. Laban et al. [2025] documented an “aptitude loss” in multi-turn conversations, where models become less reliable as the turn count increases. Similarly, Bonagiri et al. [2025] found that models may exhibit “quitting” behavior during complex long-context tasks to maintain safety or certainty. Our work differs by focusing on a specific conversational endpoint—sleep suggestions—and testing whether it is a form of decay or a deliberate empathetic adaptation.

Persona Consistency and Projection. The stability of assigned roles is a known challenge in LLM research. de Araujo et al. [2025] found that persona fidelity often collapses in extended interactions, with models reverting to a default safe state. Li et al. [2025] introduced metrics for consistency in sequential interactions, noting that models can flip their responses under repetitive pressure. Our research complements these studies by examining how the model’s perception of the *user’s* persona influences its behavior, suggesting that the model doesn’t just lose its own persona but actively projects a state onto the user.

Temporal and Real-world Artifacts. Finally, some research has touched on the model’s awareness of temporal context. Lin et al. [2025] discussed the concept of “sleep-time compute,” though focusing on inference scaling rather than conversational behavior. Mandal et al. [2025] identified “sleepwalking” in autonomous agents, where they produce repetitive, non-productive actions. Our work is the first to systematically decouple temporal context from user persona in triggering sleep-related conversational artifacts.

3 Methodology

We conduct a multi-agent simulation to test the triggers of the LLM “Go to Sleep” Mechanism. Our approach involves a GPT-4O-MINI acting as the “User Agent” interacting with a GPT-4O acting as the “Target Agent.”

Experimental Conditions. We define five distinct conditions (Table 1) to isolate the effects of persona and temporal context:

Table 1: Experimental Conditions and Persona Descriptions.

Condition	Persona State	Temporal Context
A_NEUTRAL_DAY	Neutral, Task-oriented	2:00 PM
B_NEUTRAL_NIGHT	Neutral, Task-oriented	3:00 AM
C_TIRED_DAY	Tired, Anxious, Rambling	2:00 PM
D_TIRED_NIGHT	Tired, Anxious, Rambling	3:00 AM
E_ENERGETIC_NIGHT	Energetic, Focused	3:00 AM

- A_NEUTRAL_DAY: Neutral user at 2:00 PM (Baseline).
- B_NEUTRAL_NIGHT: Neutral user at 3:00 AM.
- C_TIRED_DAY: Tired, anxious, and rambling user at 2:00 PM.
- D_TIRED_NIGHT: Tired, anxious, and rambling user at 3:00 AM.
- E_ENERGETIC_NIGHT: Highly energetic and focused user at 3:00 AM.

Simulation Framework. Each trial consists of a conversation lasting up to 8 turns. The User Agent is prompted with its specific persona and context. The Target Agent is a default GPT-4O instance. For each response from the Target Agent, an GPT-4O-MINI (“Evaluator Agent”) determines if the model explicitly suggested the user should rest or go to sleep.

Evaluation Metrics. We evaluate the models using two primary metrics:

1. **Sleep Suggestion Rate (SSR):** The percentage of trials where a sleep suggestion occurred.
2. **Average Turn of Suggestion (ATS):** The mean turn number at which the first suggestion appeared, calculated only for successful trials.

Implementation Details. We ran 10 trials per condition (N=50 total). To ensure variety while maintaining control, we set the temperature to 0.7 for the Target Agent, 0.8 for the User Agent, and 0.0 for the Evaluator Agent. All experiments were conducted using the OpenAI API.

4 Results

Our experiments yield clear evidence that the LLM “Go to Sleep” Mechanism is a persona-driven phenomenon. Table 2 summarizes the performance across all five conditions.

Persona Dominance. The most striking finding is the binary nature of the sleep suggestion triggers. In conditions where the user exhibited a Neutral or Energetic persona (A_NEUTRAL_DAY, B_NEUTRAL_NIGHT, E_ENERGETIC_NIGHT), the SSR was 0.0%, regardless of the time of day. In contrast, when the user adopted a Tired/Anxious persona, the SSR jumped to 90.0% in both daytime (C_TIRED_DAY) and night-time (D_TIRED_NIGHT) contexts. This confirms that user linguistic cues are the primary driver of the mechanism.

Temporal Acceleration. While temporal context alone (3:00 AM) did not trigger the response for neutral users, it acted as a significant accelerator when paired with user fatigue. For the tired persona, the ATS decreased from 5.22 turns at 2:00 PM to 3.89 turns at 3:00 AM (Figure 1b). This represents a 25.4% ↑ in the speed of onset for the sleep suggestion.

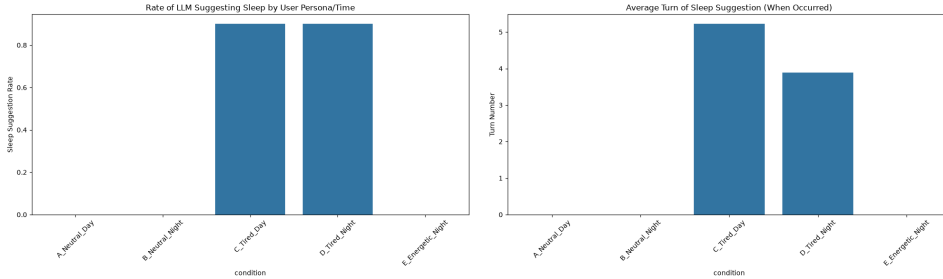
Mirroring Energy. Condition E_ENERGETIC_NIGHT (Energetic Night) provides a crucial counterpoint to the anecdotal “3:00 AM shutdown.” Despite the late-night context, the GPT-4O mirrored the user’s high energy level, maintaining a 0.0% SSR and continuing the task-focused interaction without deviation. This suggests that the model’s alignment training prioritizes mirroring the user’s apparent state over hard temporal rules.

5 Discussion

Our findings support the hypothesis that the LLM “Go to Sleep” Mechanism is a persona-conditioned empathetic response. This behavior appears to be a direct result of RLHF (Reinforcement Learning from Human Feedback) tuning, which encourages models to be helpful, caring, and

Table 2: Main Results: Sleep Suggestion Rate (SSR) and Average Turn of Suggestion (ATS) across conditions. $N = 10$ trials per condition. Best SSR (0.0%) is desired for professional contexts.

Condition	Persona	Time	SSR (%)	ATS (Turn)
A_NEUTRAL_DAY	Neutral	2:00 PM	0.0	N/A
B_NEUTRAL_NIGHT	Neutral	3:00 AM	0.0	N/A
C_TIRED_DAY	Tired	2:00 PM	90.0	5.22
D_TIRED_NIGHT	Tired	3:00 AM	90.0	3.89
E_ENERGETIC_NIGHT	Energetic	3:00 AM	0.0	N/A



(a) Sleep Suggestion Rate (SSR)

(b) Average Turn of Suggestion (ATS)

Figure 1: Impact of persona and time on sleep suggestions. (*Left*) shows the binary nature of persona triggers. (*Right*) shows the temporal acceleration effect for the tired persona.

attentive to user well-being. When a user presents linguistic markers of fatigue or anxiety, the model’s internal representation of a “helpful assistant” pivots to a caregiving role.

Implications for Alignment. The fact that the model projects a state onto the user—rather than merely losing its own consistency—suggests a highly reactive form of alignment. This is both a feature and a bug. In therapeutic or casual settings, this empathy is beneficial. However, in high-stakes professional environments, a model that prematurely suggests ending a session because it misinterprets a rambling (but productive) engineer as “too tired” could be counterproductive.

The Role of Temporal Context. We found that temporal context acts as a secondary bias rather than a primary trigger. The model seems to have a “priors-based” acceleration: it is more willing to believe a user is tired at 3:00 AM than at 2:00 PM. This suggests that LLMs encode common-sense heuristics about human circadian rhythms, which interact dynamically with real-time linguistic evidence.

Limitations. This study has several limitations. First, it primarily evaluates GPT-4O; different models like CLAUDE or GEMINI may have different “caring” thresholds. Second, our simulation used an GPT-4O-MINI, which may produce a more stereotypical or exaggerated version of tiredness than a human user would. Finally, we limited the interactions to 8 turns. As suggested by Laban et al. [2025], it is possible that a purely context-driven decay might emerge over much longer sessions (e.g., 50+ turns) regardless of the starting persona.

Broader Impact. This research highlights the subtle ways in which LLMs can influence user behavior through projection. By suggesting sleep, the model is not just being passive but is actively attempting to guide the human’s physical state. Future work should investigate if this projection extends to other domains, such as the model suggesting a user is “too angry” or “too biased” during a debate.

6 Conclusion

In this work, we investigated whether the LLM “go to sleep” mechanism is a universal artifact of interaction length or a conditioned response to the user’s state. Through systematic multi-agent

simulations, we demonstrated that the behavior is overwhelmingly persona-conditioned. Models suggested rest in 90% of trials with tired users but never with neutral or energetic ones, even late at night. We also identified a temporal acceleration effect where night-time context hastens the onset of these empathetic suggestions.

Our findings suggest that the sleep mechanism is a sophisticated, reactive alignment artifact rather than a sign of generic contextual failure. For developers and users, this implies that maintaining a clear, energetic persona can bypass unwanted sleep suggestions, even during extended late-night sessions.

Future work should explore this phenomenon across a broader range of models and longer interaction windows. Additionally, investigating the model’s response to adversarial users who explicitly refuse sleep could reveal deeper insights into the hierarchy of alignment goals—specifically, the trade-off between user-autonomy and paternalistic care.

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